Basic Introduction
Accessing Documents
General Information
STREAM
DELTA PRO
DELTA 2
DELTA 2 MAX
Smart Home Panel
Smart Plug
PowerStream Micro-inverter
Wave Air Conditioner
Glacier
Power Kits
PowerOcean
DELTA Pro Ultra
DELTA Pro 3
Smart Home Panel II
River 2 Pro
Code package download
Support

ModuleType definition

Field	Field's Type	Description
moduleType	int	1: PD 2: BMS 3: INV 4: BMS_SLAVE 5: MPPT

HTTP communication mode

Set & Get Quota

How to transfer request parameters by these two HTTP API.

PUT: /iot-open/sign/device/quota: SetCmdRequest

GET: /iot-open/sign/device/quota: GetCmdRequest, GetCmdResponse

MPPT

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

Set buzzer silent mode(0: disables the silent mode, 1: enables the silent mode)	<pre>{
	"id": 123456789,
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	"moduleType": 5,
	"operateType": "quietMode",
	"params": {
	"enabled": 1
	}
	}</pre>
		<pre>{
		"sn": "R331ZEB4ZEAL0528",
		"params": {
		"quotas": [
		"mppt.beepState"
		]
		}
		}</pre>
			<pre>{
			"code": "0",
			"message": "Success",
			"data": {
			"mppt.beepState": 1
			},
			"tid": ""
			}</pre>

Set car charger switch (1: On; 0: Off)	<pre>{ "id": 123456789,</pre>	<pre>{ "sn": "R331ZEB4ZEAL0528",</pre>	<pre>{ "code": "0",</pre>

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse
	<pre>"version": "1.0", "sn": "R331ZEB4ZEAL0528", "moduleType": 5, "operateType": "mpptCar", "params": { "enabled": 1 } }</pre>	<pre>"params": { "quotas": ["mppt.carState"] } }</pre>	<pre>"message": "Success", "data": { "mppt.carState": 1 }, "tid": "" }</pre>
Set AC discharge ("enabled" and X-Boost switch settings)	<pre>{ "id": 123456789, "version": "1.0", "sn": "R331ZEB4ZEAL0528", "moduleType": 5, "operateType": "acOutCfg", "params": { "enabled": 0, "xboost": 0, "out_voltage": 30, "out_freq": 1 } }</pre>	<pre>{ "sn": "R331ZEB4ZEAL0528", "params": { "quotas": ["mppt.cfgAcEnabled", "mppt.cfgAcXboost", "mppt.cfgAcOutVol", "mppt.cfgAcOutFreq"] } }</pre>	<pre>{ "code": "0", "message": "Success", "data": { "mppt.cfgAcOutVol": 40, "mppt.cfgAcXboost": 1, "mppt.cfgAcEnabled": 0, "mppt.cfgAcOutFreq": 50 }, "tid": "" }</pre>
AC charging settings(chgPauseFlag : 0: AC charging in normal operation, 1: AC charging paused (not saved, restored by plugging))	<pre>{ "id": 123456789, "version": "1.0", "sn": "R331ZEB4ZEAL0528",</pre>	<pre>{ "sn": "R331ZEB4ZEAL0528", "params": { "quotas": ["mppt.cfgCh</pre>	<pre>{ "code": "0", "message": "Success", "data": { "mppt.cfgCh</pre>

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

	<pre>"moduleType": 5,
	
	"operateType": "acChgCfg",
	
	"params": {
	
	"chgWatts": 100,
	
	"chgPauseFlag": 0
	}
	}</pre>
		<pre>gWatts",
		
		"mqtt.chgPauseFlag"
		]
		}
		}</pre>
			<pre>gWatts": 100
			},
			"tid": ""
			}</pre>

AC standby time when there is no load(0: never shuts down, default value: 12 x 60 mins, unit: minute)	<pre>{
	"id": 123456789,
	
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	
	"moduleType": 5,
	
	"operateType": "standbyTime",
	
	"params": {
	
	"standbyMins": 180
	}
	}</pre>
		<pre>{
		"sn": "R331ZEB4ZEAL0528",
		
		"params": {
		
		"quotas": [
		
		"mppt.acStandbyMins"
		]
		}
		}</pre>
			<pre>{
			"code": "0",
			
			"message": "Success",
			"data": {
			
			"mppt.acStandbyMins": 180
			},
			"tid": ""
			}</pre>

CAR standby duration settings(Auto shutdown when there is no load, unit: minute)	<pre>{
	"id": 123456789,
	
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	
	"moduleType": 5,
	
	"operateType": "carStandby",
	
	"params": {
	
	"standbyMin
		<pre>{
		"sn": "R331ZEB4ZEAL0528",
		
		"params": {
		
		"quotas": [
		
		"mppt.carStandbyMin"
		]
		}
		}</pre>
			<pre>{
			"code": "0",
			
			"message": "Success",
			"data": {
			
			"mppt.carStandbyMin": 240
			},
			"tid": ""
			}</pre>

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

Set 12 V DC (car charger) charging current(Maximum DC charging current (mA), range: 4000 mA–10000 mA, default value: 8000 mA)	s": 240
	}
	}
	
	{
	"id": 123456789,
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	"moduleType": 5,
	"operateType": "dcChgCfg",
	"params": {
	"dcChgCfg": 5000
	}
	}
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		"sn": "R331ZEB4ZEAL0528",
		"params": {
		"quotas": [
		"mppt.dcChgCurrent"
		]
		}
		}
			{
			"code": "0",
			"message": "Success",
			"data": {
			"mppt.dcChgCurrent": 5000
			},
			"tid": ""
			}

PD

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

Set standby time(0 for never standby; other values indicate the standby time; in minutes)	{
	"id": 123456789,
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	"moduleType": 1,
	"operateType": "standbyTime",
	"params": {
	"standbyMin": 0
	}
	}
		{
		"sn": "R331ZEB4ZEAL0528",
		"params": {
		"quotas": [
		"pd.standbyMin"
		]
		}
		}
			{
			"code": "0",
			"message": "Success",
			"data": {
			"pd.standbyMin": 60
			},
			"tid": ""
			}

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

Set DC(USB) switch(0: off, 1: on)	{
	"id": 123456789,
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	"moduleType": 1,
	"operateType": "dcOutCfg",
	"params": {
	"enabled": 0
	}
	}
		{
		"sn": "R331ZEB4ZEAL0528",
		"params": {
		"quotas": [
		"pd.dcOutState"
		]
		}
		}
			{
			"code": "0",
			"message": "Success",
			"data": {
			"pd.dcOutState": 1
			},
			"tid": ""
			}

LCD screen settings(delayOff: screen timeout, unit: seconds;brightLevel: must be set to 3; other values are invalid.)	{
	"id": 123456789,
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	"moduleType": 1,
	"operateType": "lcdCfg",
	"params": {
	"delayOff": 60,
	"brighLevel": 3
	}
	}
		{
		"sn": "R331ZEB4ZEAL0528",
		"params": {
		"quotas": [
		"pd.lcdOffSec",
		"pd.brightLevel"
		]
		}
		}
			{
			"code": "0",
			"message": "Success",
			"data": {
			"pd.brightLevel": 3,
			"pd.lcdOffSec": 60
			},
			"tid": ""
			}

Prioritize solar charging	{ "id": 123456789,	{ "sn": "R331ZCB4ZE86056P",	{ "code": "0",

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse
	"version": "1.0", "sn": "R331ZEB4ZEAL0528", "moduleType": 1, "operateType": "pvChangePrio", "params": { "pvChangeSet": 0 } }	"params": { "quotas": ["pd.pvChgPrioSet"] } }	"message": "Success", "data": { "pd.pvChgPrioSet": 0 }, "tid": "" }
Set energy management (isConfig: energy management, 0: disabled, 1: enabled; bpPowerSoc: backup reserve level; minDsgSoc: discharge limit (not in use);minChgSoc: charge limit (not in use))	{ "id": 123456789, "version": "1.0", "sn": "R331ZEB4ZEAL0528", "moduleType": 1, "operateType": "watthConfig", "params": { "isConfig": 1, "bpPowerSoc": 95, "minDsgSoc": 40, "minChgSoc": 95 } }	{ "sn": "R331ZEB4ZEAL0528", "params": { "quotas": ["pd.watchIsConfig", "pd.bpPowerSoc", "pd.minDsgSoc", "pd.minChgSoc"] } }	{ "code": "0", "message": "Success", "data": { "pd.watchIsConfig": 1, "pd.bpPowerSoc": 77 } }
Set AC always on (acAutoOutConfig: 0: disabled; 1: enabled;minAcOutSoc: minimum SoC for	{ "id": 123456789, "version": "1.0", "sn": "R331ZEB4ZE	{ "sn": "R331ZEB4ZEAL0528", "params": { "quotas": [	{ "code": "0", "message": "Success", "data": {

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

turning on "AC always on")	AL0528",
	
	"moduleType": 1,
	
	"operateType": "acAutoOutConfig",
	
	"params": {
	
	"acAutoOutConfig": 0,
	
	"minAcOutSoc": 20
	}
	}
		"pd.acAutoOutConfig",
		
		"pd.minAcoutSoc"
		]
		}
		}
			"pd.acAutoOutConfig": 0,
			
			"pd.minAcoutSoc": 20
			}
			}

BMS

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

UPS settings(UPS, upper SoC limit when charging)	{
	"id": 123456789,
	
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	
	"moduleType": 2,
	
	"operateType": "upsConfig",
	
	"params": {
	
	"maxChgSoc": 50
	}
	}
		{
		"sn": "R331ZEB4ZEAL0528",
		
		"params": {
		
		"quotas": [
		
		"bms_emsStatus.maxChargeSoc"
		]
		}
		}
			{
			"code": "0",
			
			"message": "Success",
			"data": {
			
			"bms_emsStatus.maxChargeSoc": 50
			},
			"tid": ""
			}

SOC lower limit when discharging	{
	"id": 123456789,
	
	"version": "1.0",
	"sn":
		{
		"sn": "R331ZEB4ZEAL0528",
		
		"params": {
			{
			"code": "0",
			
			"message": "Success",
			"data":

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse

	"R331ZEB4ZEAL0528",
	"moduleType": 2,
	"operateType": "dsgCfg",
	"params": {
	"minDsgSoc": 19
	}
	}
		"quotas": [
		"bms_emsStatus.minDsgSoc"
		]
		}
		}
			{
			"bms_emsStatus.minDsgSoc": 19
			},
			"tid": ""
			}

SoC that triggers EMS to turn on Smart Generator	{
	"id": 123456789,
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	"moduleType": 2,
	"operateType": "openOilSoc",
	"params": {
	"openOilSoc": 40
	}
	}
		{
		"sn": "R331ZEB4ZEAL0528",
		"params": {
		"quotas": [
		"bms_emsStatus.minOpenOilEb"
		]
		}
		}
			{
			"code": "0",
			"message": "Success",
			"data": {
			"bms_emsStatus.minOpenOilEb": 40
			},
			"tid": ""
			}

SOC that triggers EMS to turn off Smart Generator	{
	"id": 123456789,
	"version": "1.0",
	"sn": "R331ZEB4ZEAL0528",
	"moduleType": 2,
	"operateType": "closeOilSoc",
	"params": {
	"closeOilSo
		{
		"sn": "R331ZEB4ZEAL0528",
		"params": {
		"quotas": [
		"bms_emsStatus.maxCloseOilEb"
		]
		}
		}
			{
			"code": "0",
			"message": "Success",
			"data": {
			"bms_emsStatus.maxCloseOilEb": 80
			},
			"tid": ""
			}

Set Command	SetCmdRequest	GetCmdRequest	GetCmdResponse
	c": 80 } }		

GetAllQuotaResponse

How to transfer request parameters by this HTTP API.

GET: /iot-open/sign/device/quota/all: GetAllQuotaResponse

Field	Field's Type	Description
mppt.faultCode	int	Error code: byte0: mppt_fault; byte1: car_fault; byte2: dc24v_fault "swVer":"uint32", //mppt version number
mppt.dc24vState	int	DCDC 24 V switch status: 0: off; 1: on
pd.wattsInSum	int	Total input power (W) [key indicator]
bms_emsStatus.maxChargeSoc	int	Maximum charging SOC
pd.wifiVer	int	Wi-Fi version: 0x00000405 = V0.4.5
inv.cfgAcWorkMode	int	AC charging mode: 0: full power; 1: mute
bms_bmsStatus.designCap	int	Design capacity (mAh)
bms_bmsStatus.temp	int	Temperature (°C)
bms_bmsStatus.f32ShowSoc	float	Battery level SOC_float
pd.beepMode	int	BEEP mode: 0: Normal; 1: Silent
bms_bmsStatus.outputWatts	int	Output power
pd.typec2Watts	int	Type-C 2 output power (W)
inv.outputWatts	int	Discharging power (W)

Field	Field's Type	Description
bms_bmsStatus.vol	int	Voltage (mV)
inv.invOutFreq	int	Inverter output frequency (Hz): 50 or 60;
mppt.inAmp	int	PV input current (mA)
bms_emsStatus.bmsIsConnt	[int]	BMS in-place signal (3 byte): BIT0: Hardware in-place signal; BIT1: Software in-place signal
bms_emsStatus.dsgCmd	int	Discharge command
bms_bmsStatus.fullCap	int	Full capacity (mAh)
mppt.inVol	int	PV input voltage (mV)
bms_emsStatus.chgVol	int	Charging voltage
pd.icoBytes	int	ICO flag bit: BYTE0-BYTE13
inv.fanState	int	Fan status: 0: disabled; 1: Level 1; 2: Level 2; 3: Level 3
pd.usb1Watts	int	Common USB1 output power (W)
inv.cfgAcXboost	int	X-Boost switch: 0: off; 1: on
inv.outTemp	int	INV temperature (°C)
mppt.cfgAcEnabled	int	AC switch: 0: off; 1: on
mppt.res	[int]	Reserve 10 bytes
mppt.dcdc12vAmp	int	DC 12 V 30 A output current (mA)
inv.invOutVol	int	Inverter actual output voltage (mV)
pd.extRj45Port	int	RJ45 port: 0: NULL; 1: RC(BLE_CTL)
bms_emsStatus.chgAmp	int	Charging current
bms_bmsStatus.errCode	int	Global error code
inv.inputWatts	int	Charging power (W)

Field	Field's Type	Description
mppt.cfgAcOutVol	int	Output voltage configured for the inverter (V)
bms_emsStatus.chgState	int	Charging status
bms_bmsStatus.inputWatts	int	Input power [key indicator]
bms_emsStatus.openBmsIdx	int	Battery pack enabling status
pd.typec2Temp	int	Type-C 2 temperature (℃)
pd.carUsedTime	int	CAR use time (s)
pd.typec1Watts	int	Type-C 1 output power (W) [key indicator]
pd.chgDsgState	int	Charging/discharging status on screen: 1: discharging; 2: charging
inv.chgPauseFlag	int	PV charging pause flag bit: 1: charging stopped
inv.acInFreq	int	Inverter input frequency (Hz)
mppt.carStandbyMin	int	Auto shutdown when there is no load: 0: Never shut down; default value: 12*60mins; unit: min [key indicator]
pd.soc	int	Display SOC [key indicator]
inv.invOutAmp	int	Inverter output current (mA)
bms_emsStatus.fanLevel	int	Fan level
mppt.cfgAcOutFreq	int	Output frequency configured for the inverter (Hz)
mppt.carOutVol	int	Car charger output voltage (mV)
inv.dcInVol	int	DC input voltage (mV)
pd.ext3p8Port	int	3+8 ports: 0: NULL; 1: CC; 2: PR; 3: SP (BC)

Field	Field's Type	Description
mppt.cfgAcXboost	int	X-Boost switch: 1: On; 0: Off [key indicator]
pd.dsgPowerDC	int	Cumulative DC discharge capacity (Wh)
inv.dcInAmp	int	DC input current (mA)
pd.typecUsedTime	int	Type-C use time (s)
mppt.carOutAmp	int	Car charger output current (mA)
bms_bmsStatus.remainCap	int	Remaining capacity (mAh)
pd.brightLevel	int	LCD brightness level: 0-3 levels
inv.dcInTemp	int	DC temperature (℃)
bms_emsStatus.maxAvailNum	int	Maximum available quantity
bms_emsStatus.maxCloseOilEb	int	Disable SOC of Smart Generator [key indicator]
mppt.dcdc12vVol	int	DC 12 V 30 A output voltage (mV)
pd.wattsOutSum	int	Total output power (W) [key indicator]
mppt.scrStandbyMin	int	Auto shutdown when there is no load: 0: never shut down, default value: 12 x 60 mins, unit: minutes
mppt.carOutWatts	int	Car charger output power (W)
bms_emsStatus.f32LcdShowSoc	float	SoC value displayed on LCD - used for displaying SOC with decimal point [key indicator]
inv.cfgAcEnabled	int	AC switch: 0: off; 1: on
mppt.outVol	int	PV output voltage (mV)
bms_emsStatus.paraVolMax	int	Maximum voltage when two devices work in parallel

Field	Field's Type	Description
mppt.chgType	int	Actual charging type: 0: null; 1: Adapter (adapter/DC power); 2: MPPT (solar energy); 3: AC (grid charging); 4: Gas (petrol and electricity); 5: Wind (wind power) [key indicator]
bms_bmsStatus.bmsFault	int	BMS permanent fault
inv.acDipSwitch	int	AC fast/slow charging dip switch: 0: unknown; 1: fast charging mode; 2: slow charging mode
mppt.cfgChgType	int	Configured charging type, which is valid when xt60_chg_type is 0: 0: Auto; 1: MPPT; 2: Adapter
bms_emsStatus.paraVolMin	int	Minimum voltage when two devices work in parallel
inv.standbyMins	int	Auto shutdown when there is no load: 0: never shut down, default value: 12 x 60 mins, unit: minutes
inv.dischargeType	int	Discharging type: 1: AC discharging; 2: PR; 3: BC
pd.carState	int	CAR button status: 0: OFF; 1: ON
mppt.chgState	int	Charging status: 0: Off; 1: Charging; 2: Standby (during AC charging, DC charging stops)
pd.invUsedTime	int	Inverter use time (s)
bms_emsStatus.bmsWarState	int	BMS warning state: bit0: hi_temp; bit1: low_temp; bit2: overload; bit3: chg_flag
mppt.carState	int	Car charger switch status: 0: Off; 1: On [key indicator]

Field	Field's Type	Description
mppt.dischargeType	int	Discharging type: 1: AC discharging; 2: PR; 3: BC
pd.typec1Temp	int	Type-C 1 temperature (℃)
pd.dcInUsedTime	int	DC charging time (s)
bms_bmsStatus.sysVer	int	System version
mppt.carTemp	int	Car charging temperature (℃)
pd.model	int	Product model: see ems_model enumeration for details
bms_bmsStatus.minCellVol	int	Minimum cell voltage
mppt.outWatts	int	PV output power (W)
pd.wifiAutoRcvy	int	1: Wi-Fi automatically restores the last usage mode (STA/AP) after being powered on; 0: Default mode (STA)
pd.remainTime	int	Available time (min): >0: Time remaining before full charging; <0: Time remaining before full discharge [key indicator]
bms_bmsStatus.maxCellVol	int	Maximum cell voltage
bms_bmsStatus.type	int	BMS type: 1: lithium battery; 2: oil-powered
bms_bmsStatus.maxCellTemp	int	Maximum cell temperature
mppt.outAmp	int	PV output current (mA)
pd.ext4p8Port	int	4+8 ports; only supports left port status identification: 0: NULL; 1: Extra battery; 2: Smart generator
bms_bmsStatus.cellId	int	Cell material LI/LFP/LA, battery capacity type: 1: 2.5 Ah per battery; 2: 2 Ah per battery
bms_bmsStatus.minMosTemp	int	Minimum MOS temperature

Field	Field's Type	Description
mppt.chgPauseFlag	int	PV charging pause flag bit: 1: charging stopped
bms_emsStatus.minOpenOilEb	int	Enable SOC of Smart Generator [key indicator]
mppt.acStandbyMins	int	Auto shutdown when there is no load: 0: Never shut down; default: 12*60mins; unit: min [key indicator]
mppt.powStandbyMin	int	Auto shutdown when there is no load: 0: Never shut down; default: 12*60mins; unit: min
inv.chargerType	int	Charger type: 1: AC charging; 2: DC adapter charging; 3: solar charging; 4: CC; 5: BC
pd.chgSunPower	int	Cumulative solar charge capacity (Wh)
pd.carTemp	int	CAR temperature (℃)
bms_bmsStatus.bqSysStatReg	int	BQ hardware protection register
inv.acInAmp	int	Inverter input current (mA)
bms_emsStatus.chgRemainTime	int	Remaining charging time (min)
pd.dsgPowerAC	int	Cumulative AC power discharged (Wh)
bms_bmsStatus.openBmsIdx	int	Battery pack enabling status
pd.qcUsb2Watts	int	qc_usb2 output power (W)
pd.wireWatts	int	Wireless charging output power (W); reserved and not in use
bms_bmsStatus.num	int	BMS number: 0–2
pd.chgPowerAC	int	Cumulative AC charge (wall socket) (Wh)

Field	Field's Type	Description
pd.lcdOffSec	int	LCD screen timeout: 0: Always on [key indicator]
bms_bmsStatus.soc	int	Battery level
inv.acInVol	int	Inverter input voltage (mV)
inv.FastChgWatts	int	Maximum charging power for AC fast charging (W)
mppt.beepState	int	Buzzer status: 0: Default; 1: Silent mode [Key Indicators]
pd.sysVer	int	System version: 0x0102002F = V1.2.0.47
mppt.dc24vTemp	int	DCDC 24 V temperature (℃)
bms_emsStatus.chgCmd	int	Charge command
bms_bmsStatus.tagChgAmp	int	Target charging current
bms_bmsStatus.maxMosTemp	int	Maximum MOS temperature
mppt.cfgChgWatts	int	AC maximum charging power (W) [key indicator]
pd.qcUsb1Watts	int	qc_usb1 output power (W)
pd.reserved	int	Reserve 2 bytes
bms_bmsStatus.minCellTemp	int	Minimum cell temperature
pd.chgPowerDC	int	Cumulative DC charge (adapter) (Wh)
pd.standbyMin	int	Standby auto shutdown time (min): 0: Never standby; maximum 5999 minutes (99 hours and 59 minutes) [key indicator]
mppt.swVer	int	MPPT version number
inv.cfgAcOutFreq	int	Configured output frequency for inverter (Hz) [key indicator]

Field	Field's Type	Description
mppt.x60ChgType	int	XT60 paddle status: 0: Not detected; 1: MPPT; 2: Adapter
bms_bmsStatus.soh	int	Health status
inv.errCode	int	Global error code
bms_emsStatus.openUpsFlag	int	UPS mode enable flag
bms_emsStatus.minDsgSoc	int	Minimum discharge SoC [key indicator]
mppt.dcdc12vWatts	int	DC 12 V 30 A output power (W)
pd.usbqcUsedTime	int	USB QC use time (s)
pd.dcOutState	int	DC button status: 0: OFF; 1: ON [key indicator]
bms_bmsStatus.remainTime	int	Time remaining
bms_emsStatus.emsIsNormalFlag	int	0:sleep 1:normal
mppt.inWatts	int	PV input power (W) [key indicator]
pd.usbUsedTime	int	USB use time (s)
pd.mpptUsedTime	int	MPPT use time (s)
inv.reserved	[int]	Reserve 8 bytes
mppt.mpptTemp	int	MPPT temperature (℃)
pd.wifiRssi	int	Wi-Fi signal strength
bms_bmsStatus.amp	int	Current (mA)
inv.invType	int	PSDR model code (corresponds to dip Switch and high-low voltage switch)
bms_emsStatus.lcdShowSoc	int	SoC value displayed on LCD [key indicator]
inv.cfgAcOutVol	int	Output voltage configured for the inverter (V)

Field	Field's Type	Description
bms_emsStatus.bmsModel	int	BMS model [key indicator]
pd.errCode	int	Global error code
pd.carWatts	int	Car output power (W) [key indicator]
pd.usb2Watts	int	Normal USB2 output power (W)
mppt.dcChgCurrent	int	DC maximum charging current (mA) [key indicator]
bms_emsStatus.dsgRemainTime	int	Remaining discharging time (min)
inv.sysVer	int	System version

Example


```
{
    "code": "0",
    "message": "Success",
    "data": {
        "pd.wattsInSum": 0,
        "bms_emsStatus.maxChargeSoc": 83,
        "pd.wifiVer": 0,
        "inv.cfgAcWorkMode": 0,
        "bms_bmsStatus.designCap": 20000,
        "bms_bmsStatus.temp": 32,
        "bms_bmsStatus.f32ShowSoc": 79.4,
        "pd.beepMode": 0,
        "bms_bmsStatus.outputWatts": 0,
        "pd.typec2Watts": 0,
        "inv.outputWatts": 0,
        "bms_bmsStatus.vol": 53252,
        "inv.invOutFreq": 0,
        "mppt.inAmp": 1,
        "bms_emsStatus.chgAmp": 25000,
        "bms_bmsStatus.errCode": 0,
        "inv.inputWatts": 0,
        "mppt.cfgAcOutVol": 0,
        "bms_emsStatus.chgState": 1,
        "bms_bmsStatus.inputWatts": 0,
        "bms_emsStatus.openBmsIdx": 1,
        ...
    }
}
```


MQTT communication mode

Set & Set Reply

Usage of Topic	Topic	From	To
Set device function	/open/${certificateAccount}/${sn}/set	app	device

Usage of Topic	Topic	From	To
Reply to the set result	/open/${certificateAccount}/${sn}/set_reply	device	app

Full Format Example

Set Data Format

```
{
    "id": 123,
    "version": "1.0",
    "moduleType": 1,
    "operateType": "standbyTime",
    "params": {
        "standbyMin": 60
    }
}
```

Set Reply Data Format

```
{
    "id": 123,
    "version": "1.0",
    "moduleType": 1,
    "time": 2557051,
    "code": 0,
    "operateType": "standbyTime",
    "data": {
        "ack": 0
    }
}
```

ParamInfo Message Format

MPPT

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Indicator

Set buzzer silent mode(0: Disable silent mode; 1: Enable silent mode)	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 5,
	"operateType": "quietMode",
	"params": {
	"enabled": 1
	}
	}
		{
		"code": "0",
		"moduleType": 5,
		"data": {
		"ack": 0
		},
		"operateType": "quietMode",
		"version": "1.0",
		"id": 123456789,
		"time": 12508887
		}
			quietMode

Set car charger switch(1: On; 0: Off)	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 5,
	"operateType": "mpptCar",
	"params": {
	"enalbed": 1
	}
	}
		{
		"code": "0",
		"moduleType": 5,
		"data": {
		"ack": 1
		},
		"operateType": "mpptCar",
		"version": "1.0",
		"id": 123456789,
		"time": 12755887
		}
			mpptCar

Set AC discharge("enabled" and X-Boost switch settings)	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 5,
	"operateTyp
		{
		"code": "0",
		"moduleType": 5,
		"data": {
		"ack": 0
		},
			acOutCfg
			xboost
			out_voltage
			out_freq

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Indicator
	e": "acOutCfg", "params": { "enabled": 0, "xboost": 0, "out_voltage": 30, "out_freq": 1 } }	"operateType": "acOutCfg", "version": "1.0", "id": 123456789, "time": 13010777 }	
AC charging settings(chgPauseFlag: 0: AC charging in normal operation, 1: AC charging paused (not saved, restored by plugging))	{ "id": 123456789, "version": "1.0", "moduleType": 5, "operateType": "acChgCfg", "params": { "chgWatts": 120, "chgPauseFlag": 0 } }	{ "code": "0", "moduleType": 5, "data": { "ack": 0 }, "operateType": "acChgCfg", "version": "1.0", "id": 123456789, "time": 13137427 }	chgWatts chgPauseFlag
AC standby time when there is no load(0: never shuts down, default value: 12 x 60 mins, unit: minute)	{ "id": 123456789, "version": "1.0", "moduleType": 5, "operateType": "standbyTime", "params": { "standbyMins": 240 }	{ "code": "0", "moduleType": 5, "data": { "ack": 0 }, "operateType": "standbyTime", "version": "1.0", "id":	standbyTime

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Indicator
	}	123456789, "time": 13212227 }	

CAR standby duration settings(Auto shutdown when there is no load, unit: minute)	{
	"id":
	123456789,
	"version":
	"1.0",
	"moduleType
	": 5,
	"operateTyp
	e":
	"carStandby
	",
	"params": {
	"standbyMin
	s": 300
	}
	}
		{
		"code":
		"0",
		"moduleType
		": 5,
		"data":
		{
		"ack": 0
		},
		"operateTyp
		e":
		"carStandby
		",
		"version":
		"1.0",
		"id":
		123456789,
		"time":
		13408207
		}
			carStandby

Set 12 V DC (car charger) charging current(Maximum DC charging current (mA), range: 4000 mA–10000 mA, default value: 8000 mA)	{
	"id":
	123456789,
	"version":
	"1.0",
	"moduleType
	": 5,
	"operateTyp
	e":
	"dcChgCfg",
	"params": {
	"dcChgCfg":
	6000
	}
	}
		{
		"code":
		"0",
		"moduleType
		": 5,
		"data":
		{
		"ack": 1
		},
		"operateTyp
		e":
		"dcChgCfg",
		"version":
		"1.0",
		"id":
		123456789,
		"time":
		13508217
		}
			dcChgCfg

PD

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Observation Indicator

Set standby time(0 for never standby; other values indicate the standby time; in minutes)	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 1,
	"operateType": "standbyTime",
	"params": {
	"standbyMin": 10
	}
	}
		{
		"code": "0",
		"moduleType": 1,
		"data": {
		"ack": 0
		},
		"operateType": "standbyTime",
		"version": "1.0",
		"id": 123456789,
		"time": 13988947
		}
			standbyMin

Set DC(USB) switch(0: off, 1: on)	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 1,
	"operateType": "dcOutCfg",
	"params": {
	"enabled": 1
	}
	}
		{
		"code": "0",
		"moduleType": 1,
		"data": {
		"ack": 0
		},
		"operateType": "dcOutCfg",
		"version": "1.0",
		"id": 123456789,
		"time": 14136757
		}
			dcOutCfg

LCD screen(delayOff: screen timeout, unit: seconds;brightLevel: must be set to 3; other values are invalid.)	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 1,
	"operateTyp
		{
		"code": "0",
		"moduleType": 1,
		"data": {
		"ack": 0
		},
			delayOff
			brighLevel

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Observation Indicator
	e": "lcdCfg", "params": { "delayOff": 60, "brighLevel": 1 } }	"operateType": "lcdCfg", "version": "1.0", "id": 123456789, "time": 14277817 }	
Prioritize solar charging	{ "id": 123456789, "version": "1.0", "moduleType": 1, "operateType": "pvChangePrio", "params": { "pvChangeSet": 1 } }	{ "code": "0", "moduleType": 1, "data": { "ack": 0 }, "operateType": "pvChangePrio", "version": "1.0", "id": 123456789, "time": 2859321 }	pd.pvChgPrioSet
Set energy management(isConfig: energy management configuration, 0: disabled, 1: enabled; bpPowerSoc: backup reserve level; minDsgSoc: discharge limit (not in use);minChgSoc: charge limit (not in use))	{ "id": 123456789, "version": "1.0", "moduleType": 1, "operateType": "watthConfig", "params": { "isConfig": 1, "bpPowerSoc": 95, "minDsgSoc": 255,	{ "code": "0", "moduleType": 1, "data": { "ack": 0 }, "operateType": "watthConfig", "version": "1.0", "id": 123456789, "time": 4915011 }	pd.watchIsConfig pd.bpPowerSoc

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Observation Indicator
	"minChgSoc": 255 } }		

Set AC always on (acAutoOutConfig: 0: disabled; 1: enabled;minAcOutSoc: minimum SoC for turning on "AC always on")	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 1,
	"operateType": "acAutoOutConfig",
	"params": {
	"acAutoOutConfig": 0,
	"minAcOutSoc": 20
	}
	}
		{
		"code": "0",
		"moduleType": 1,
		"data": {
		"ack": 0
		},
		"operateType": "acAutoOutConfig",
		"version": "1.0",
		"id": 123456789,
		"time": 6675061
		}
			pd.acAutoOutConfig
			pd.minAcoutSoc

BMS

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Observation Indicator

UPS settings(UPS, upper SoC limit when charging)	{
	"id": 123456789,
	"version": "1.0",
	"moduleType": 2,
	"operateType": "upsConfig",
	"params": {
	"maxChgSoc": 50
	}
	}
		{
		"code": "0",
		"moduleType": 2,
		"data": {
		"ack": 0
		},
		"operateType": "upsConfig",
		"version": "1.0",
		"id": 123456789,
			maxChgSoc

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Observation Indicator
		<pre>"time": 77047697 }</pre>	
SOC lower limit when discharging	<pre>{ "id": 123456789, "version": "1.0", "moduleType": 2, "operateType": "dsgCfg", "params": { "minDsgSoc": 20 } }</pre>	<pre>{ "code": "0", "moduleType": 2, "data": { "ack": 0 }, "operateType": "dsgCfg", "version": "1.0", "id": 123456789, "time": 77117117 }</pre>	minDsgSoc
SoC that triggers EMS to turn on Smart Generator	<pre>{ "id": 123456789, "version": "1.0", "moduleType": 2, "operateType": "openOilSoc", "params": { "openOilSoc": 40 } }</pre>	<pre>{ "code": "0", "moduleType": 2, "data": { "ack": 0 }, "operateType": "openOilSoc", "version": "1.0", "id": 123456789, "time": 77215347 }</pre>	openOilSoc
SoC that triggers EMS to turn off Smart Generator	<pre>{ "id": 123456789,</pre>	<pre>{ "code": "0",</pre>	closeOilSoc

Set Command	SetTopic's ParamInfo	SetReplyTopic's ParamInfo	Observation Indicator
	<pre>"version": "1.0", "sn": "R331ZEB4ZEAL0528", "moduleType": 2, "operateType": "closeOilSoc", "params": { "closeOilSoc": 90 } }</pre>	<pre>"moduleType": 2, "data": { "ack": 0 }, "operateType": "closeOilSoc", "version": "1.0", "id": 123456789, "time": 77268677 }</pre>	

Report Device Quota

Usage of Topic	Topic	From	To
Report device quota	/open/${certificateAccount}/${sn}/quota	device	app

Please see HTTP GetAllQuotaResponse and Set & Set Reply for the fields definition.

Example

PD


```
{
    "id": 73607317,
    "version": "1.0",
    "time": 73607327,
    "moduleType": 1,
    "typeCode": "pdStatus",
    "params": {
        "model": 80,
        "errCode": 0,
        "sysVer": 16908376,
        "wifiVer": 0,
        "wifiAutoRcvy": 0,
        "soc": 79,
        "wattsOutSum": 0,
        "wattsInSum": 0,
        "remainTime": -5939,
        "beepMode": 0,
        "dcOutState": 0,
        "usb1Watts": 0,
        "usb2Watts": 0,
        "qcUsb1Watts": 0,
        "qcUsb2Watts": 0,
        "typec1Watts": 0,
        "typec2Watts": 0,
        "typec1Temp": 0,
        "typec2Temp": 0,
        "carState": 0,
```

```
        "carWatts": 0,
        "carTemp": 38,
        "standbyMin": 0,
        "lcdOffSec": 0,
        "brightLevel": 3,
        "chgPowerDC": 0,
        "chgSunPower": 0,
        "chgPowerAC": 0,
        "dsgPowerDC": 0,
        "dsgPowerAC": 0,
        "usbUsedTime": 0,
        "usbqcUsedTime": 0,
        "typecUsedTime": 0,
        "carUsedTime": 0,
        "invUsedTime": 0,
        "dcInUsedTime": 0,
        "mpptUsedTime": 0,
        "reserved": [
            0,
            0
        ],
        "extRj45Port": 0,
        "ext3p8Port": 0,
        "ext4p8Port": 0,
        "chgDsgState": 0,
        "wifiRssi": 0,
        "wireWatts": 0
    }
}
```


BMS-bmsStatus


```
{
    "id": 536890212,
    "version": "1.0",
    "time": 73628997,
    "moduleType": 2,
    "typeCode": "bmsStatus",
    "params": {
        "vol": 53251,
        "amp": -138,
        "f32ShowSoc": 79.3
    }
}
```


BMS-emsStatus


```
{
    "id": 536890212,
    "version": "1.0",
    "time": 73607827,
    "moduleType": 2,
    "typeCode": "emsStatus",
    "params": {
        "chgState": 1,
        "chgCmd": 1,
        "dsgCmd": 1,
        "chgVol": 55251,
        "chgAmp": 25000,
        "fanLevel": 0,
        "maxChargeSoc": 83,
        "bmsModel": 13,
        "lcdShowSoc": 79,
        "openUpsFlag": 1,
        "bmsWarState": 0,
        "chgRemainTime": 5939,
        "dsgRemainTime": 5939,
        "emsIsNormalFlag": 1,
        "f32LcdShowSoc": 79.3,
        "bmsIsConnt": [
            3,
            3,
            0
        ],
        "maxAvailNum": 1,
        "openBmsIdx": 1,
        "paraVolMin": 52259,
        "paraVolMax": 54259,
```

```
        "minDsgSoc": 22,
        "minOpenOilEb": 40,
        "maxCloseOilEb": 90
    }
}
```


MPPT


```
{
    "id": 73607607,
    "version": "1.0",
    "time": 73607617,
    "moduleType": 5,
    "typeCode": "mpptStatus",
    "params": {
        "faultCode": 0,
        "swVer": 83951779,
        "inVol": 1522,
        "inAmp": 1,
        "inWatts": 0,
        "outVol": 52821,
        "outAmp": 0,
        "outWatts": 0,
        "mpptTemp": 36,
        "x60ChgType": 0,
        "cfgChgType": 0,
        "chgType": 0,
        "chgState": 0,
        "dcdc12vVol": 0,
        "dcdc12vAmp": 0,
        "dcdc12vWatts": 0,
        "carOutVol": 51,
        "carOutAmp": 0,
        "carOutWatts": 0,
        "carTemp": 38,
        "carState": 0,
        "dc24vTemp": 0,
        "dc24vState": 0,
        "chgPauseFlag": 0,
        "dcChgCurrent": 6000,
        "beepState": 0,
        "cfgAcEnabled": 0,
        "cfgAcXboost": 1,
        "cfgAcOutVol": 0,
        "cfgAcOutFreq": 50,
        "cfgChgWatts": 120,
        "acStandbyMins": 0,
        "dischargeType": 0,
        "carStandbyMin": 0,
        "powStandbyMin": 720,
        "scrStandbyMin": 120,
        "res": [
            0,
            0,
            0,
            0,
            0
        ]
    }
}
```


INV


```
{
    "id": 6910941,
    "version": "1.0",
    "time": 6910941,
    "moduleType": 3,
    "needAck": 0,
    "typeCode": "invStatus",
    "params": {
        "errCode": 0,
        "sysVer": 50331766,
        "chargerType": 0,
        "inputWatts": 0,
        "outputWatts": 0,
        "invType": 0,
        "invOutVol": 229104,
```

```
            "invOutAmp": 0,
            "invOutFreq": 50,
            "acInVol": 0,
            "acInAmp": 0,
            "acInFreq": 0,
            "outTemp": 39,
            "dcInVol": 0,
            "dcInAmp": 0,
            "dcInTemp": 0,
            "fanState": 0,
            "cfgAcEnabled": 1,
            "cfgAcXboost": 0,
            "cfgAcOutVol": 230000,
            "cfgAcOutFreq": 1,
            "cfgAcWorkMode": 0,
            "chgPauseFlag": 0,
            "acDipSwitch": 0,
            "FastChgWatts": 0,
            "SlowChgWatts": 0,
            "standbyMins": 720,
            "dischargeType": 0,
            "reserved": [
                0,
                0,
                0,
                0,
                0,
                0,
                0,
                0
            ]
        }
    }
```


Report Device status

Usage of Topic	Topic	From	To
Report Device status	/open/${certificateAccount}/${sn}/status	device	app

Message Format


```
    {
        "id": "123456789",
        "version": "1.0",
        "timestamp": 1634841971000,
        "params": {
            "status": 0
        }
    }
```


Field

Field	Field Type	Description
status	int	Device online or not 0: No, 1: Yes

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YES NO

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